

Nanjie Yao

(+86) 199-0581-7655 | nanjieyao@gmail.com | jackiemin233.github.io | [Google Scholar](#)

EDUCATION

The Hong Kong University of Science and Technology (Guangzhou) Guangzhou, China
Ph.D. in Artificial Intelligence Sept. 2026 – Jun. 2030 (*Expected*)

- Supervisor: Prof. Hao Wang
- Research Interests: 3DV, LLM Agent

Zhejiang University of Technology Hangzhou, China
B.S. in Network Engineering Sept. 2021 – Jun. 2025

- GPA: 3.71/4.0 (Rank: 4/53)
- Supervisor: Prof. Sixian Chan
- Honor Graduate of Jianxing Honors College

University of California - Irvine Irvine, USA
Visiting Student Sept. 2023 – Dec. 2023

- GPA: 4.0/4.0
- Academic Study Abroad Program
- Enroll courses in AI, computer system security, data science and participate in a research project.

WORK EXPERIENCE

Algorithm Engineer - Intern Shanghai, China
Horizon Robotics Aug. 2026 – Jan. 2027

Research Topic: Video Generation; World Model; Vision Language Action

Research Assistant Guangzhou, China
HKUST (GZ) Apr. 2024 – Aug. 2026

Supervisor: Prof. Hao Wang

Research Topic: 3D Reconstruction; Human Animation; Human Pose Estimation

Remote Intern Texas, USA
TA&MU Aug. 2025 – Sept. 2025

Supervisor: Prof. Zhengzhong Tu

Research Topic: Human Video Generation

Research Assistant (Part-time) Hangzhou, China
Zhejiang University of Technology Apr. 2024 – Sept. 2024

Supervisor: Prof. Zhenyu Wen & Prof. Yiming Wu

Research Topic: Auto-ML; Neural Backdoor Detection and Mitigation

Research Assistant - Intern Hangzhou, China
Zhejiang Lab Dec. 2023 – Apr. 2024

Supervisor: Prof. Feng Lin

Research Topic: 3D Reconstruction

PUBLICATIONS

(* denotes equal contribution)

MotionGRPO: Overcoming Low Intra-Group Diversity in GRPO-Based Egocentric Motion Recovery

- International Conference on Machine Learning (ICML), 2026
- Nanjie Yao, Junlong Ren, Wenhao Shen, Hao Wang

MultiGO++: Monocular 3D Clothed Human Reconstruction via Geometry-Texture Collaboration

- IEEE Transactions on Visualization and Computer Graphics (IEEE TVCG), 2026
- Nanjie Yao*, Gangjian Zhang*, Wenhao Shen, Jian Shu, Hao Wang

MultiGO: Towards Multi-level Geometry Learning for Monocular 3D Textured Human Reconstruction

- IEEE Conference on Computer Vision and Pattern Recognition (IEEE CVPR), 2025
- Gangjian Zhang*, **Nanjie Yao***, Shunsi Zhang, Hanfeng Zhao, Guoliang Pang, Jian Shu, Hao Wang

FastAnimate: Towards Learnable Template Construction and Pose Deformation for Fast 3D Human Avatar Animation

- Annual AAAI Conference on Artificial Intelligence (AAAI), 2026
- Jian Shu*, **Nanjie Yao***, Gangjian Zhang, Junlong Ren, Yu Feng, Hao Wang

End-to-End Model Generation with Large Language Models for Adaptive IoT Application Deployment

- IEEE/ACM International Conference on Software Engineering (IEEE/ACM ICSE), 2026
- Zhenyu Wen, Jintao Feng, **Nanjie Yao**, Di Wu, Cong Wang, Shibo He

CaM-Wolf: Causal-Aware Multimodal Agents for Social Deduction Games

- ACM International Conference on Multimedia (ACM MM), 2026
- Zheng Zhang, **Nanjie Yao**, Jiarui He, Deheng Ye, Peilin Zhao, and Hao Wang

Horizon-Stream: Long-Horizon Attention for Streaming 3D Reconstruction

- The Fortieth Annual Conference on Neural Information Processing Systems (NeurIPS), 2026
- Chong Cheng, Peilin Tao, **Nanjie Yao**, Guanzhi Ding, Xianda Chen, yuansen du, Xiaoyang Guo, Wei Yin, Weiqiang Ren, Qian Zhang, Zhengqing Chen, Hao Wang

SAT: Supervisor Regularization and Animation Augmentation for Two-process Monocular Texture 3D Human Reconstruction

- ACM International Conference on Multimedia (ACM MM), 2025
- Gangjian Zhang, Jian Shu, **Nanjie Yao**, Hao Wang

SMPL Normal Map Is All You Need for Single-view Textured Human Reconstruction

- IEEE International Conference on Multimedia & Expo (IEEE ICME, Oral Presentation), 2025
- Wenhao Shen, Gangjian Zhang, Jianfeng Zhang, Yu Feng, **Nanjie Yao**, Xuanmeng Zhang, Hao Wang

Human Multi-View Synthesis from a Single-View Model: Transferred Body and Region Representations

- IEEE International Conference on Multimedia & Expo (IEEE ICME), 2026
- Yu Feng, Shunsi Zhang, Jian Shu, **Nanjie Yao**, Gangjian Zhang, Hanfeng Zhao, Lie Xu, Chi Zhang, Hao Wang

PREPRINT AND MANUSCRIPTS

DeepSanitizer: Data-free Backdoor Removal in Deep Neural Networks

- Submitted to IEEE Transactions on Dependable and Secure Computing (IEEE TDSC), 2025 (Under Review)
- Yiming Wu*, **Nanjie Yao***, Chong Fu, Chenghui Shi, Tiejun Wu, Shouling Ji

Routed Frequency Factorization for Human Motion Estimation from Sparse VR Signals

- Submitted to Annual AAAI Conference on Artificial Intelligence (AAAI), 2027
- Wenhao Shen*, **Nanjie Yao***, Guoshen Lin, Hao Wang

DUET4D: Self-Corrective Coupled Guidance for Sparse-View 4D Reconstruction

- Submitted to Annual AAAI Conference on Artificial Intelligence (AAAI), 2027
- Beizhen Zhao*, **Nanjie Yao***, Boyi Fu, Qingsong Yan, Haofan Ren, Kaiyong Zhao, Hao Wang

TAG-Edit3D: Learning Transferable Edit Directions with Adaptive Geometric Guidance for 3D Scene Editing

- Submitted to Annual AAAI Conference on Artificial Intelligence (AAAI), 2027
- Guanzhi Ding*, **Nanjie Yao***, Beizhen Zhao, Sicheng Yu, Qingsong Yan, Haofan Ren, Kaiyong Zhao, Hao Wang

MIRROR: Motion-Internalized Reasoning with Reciprocal On-Policy Reinforcement for Unified Motion Generation and Understanding

- Submitted to Annual AAAI Conference on Artificial Intelligence (AAAI), 2027
- Wenhao Shen, Hao Wang, **Nanjie Yao**, Shanshan Zhao, Qing-Guo Chen, Weihua Luo, Guosheng Lin

EvolveNav: Proactive Preflection and Self-Evolving Memory for Zero-Shot Object Goal Navigation

- Submitted to ACM International Conference on Multimedia (ACM MM), 2026
- Qi Chai, Wenhao Shen, **Nanjie Yao**, Yue Xia, Kaiyong Zhao, Jie Ma, Guosheng Lin, Hao Wang

Orchestrating LLMs with Specialized Models: A Survey on Heterogeneous Multi-Model Agents

- Submitted to ACM Computing Survey (ACM CSUR), 2026
- Zheng Zhang*, **Nanjie Yao***, Yu Feng, Qi Chai, Liu Liu, Deheng Ye, Peilin Zhao, Xiangxin Zhou, Hao Wang, Hui Xiong

World Models for Embodied Intelligence: From Plausible to Controllable to Actionable

- Submitted to IEEE Transactions on Pattern Recognition and Machine Intelligence (IEEE TPAMI), 2026
- **Nanjie Yao**, Hao Wang, Chong Cheng, Zhikang Chen, Wenzhe Li, Jiafei Lyu, Li Shen, Peilin Zhao, Zongqing Lu, Gao Huang, Steven Hoi, Dacheng Tao, Deheng Ye

TECHNICAL SKILLS

Program Languages: Python, Bash, C/C++, Latex, Markdown, Java, Html, CSS, SQL
Frameworks: Pytorch, Numpy, Matplotlib, Pytorch3D, Open3D, Springboot, Mybatis
Developer Tools: Linux, Docker, Blender, MeshLab, VS Code

AWARDS & HONORS

Excellent Undergraduate Dissertation Project	Jun. 2025
Honor Graduate from Jianxing Honors College	Jun. 2025
Zhejiang Provincial Government Scholarship (Top 3%)	Sept. 2024
1st Class Scholarship for Outstanding Students (Top 1%)	Sept. 2024
1st Class Scholarship for Academic Record (Top 1%)	Sept. 2024
Huawei Intelligent Base Scholarship (6000 CNY)	Dec. 2023
2nd Class Scholarship for Outstanding Students (Top 8%)	Sept. 2023
2nd Class Scholarship for Academic Record (Top 8%)	Sept. 2023
2nd Prize in the National Information Security Competition for Chinese College Students	Jul. 2023
2nd Prize of the National Innovation and Entrepreneurship Competition of China	Jun. 2023
3rd Prize in the China RoboCup (Drone Track)	Dec. 2022

PATENTS

A 3D Organ Medical Image Segmentation System and Method Based on Paddle-Seg Deep Learning Framework – Chinese Software Copyright	2024
--	------

COMMUNITY SERVICES

Reviewer: ICASSP 2026, 2027; ICME 2025; TMM 2026